
Chapter 36

Property Rights in the New Economy

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As e-commerce expanded, the legal definition of intellectual property expanded as well, spurring increased intellectual property activity in new industries. The legal definition of invention, for example, has expanded to include computer software and methods of doing business on the Internet. Whether these developments are good for the new economy has generated much controversy. This chapter provides some background material and issues to consider. The following questions are addressed: (1) How are new economy technologies (such as software and Internet innovations) protected? (2) What is the state of intellectual property laws pertaining to these new technologies? How significant is intellectual property activity in these new industries? (3) Are intellectual property rights (IPRs) appropriate for new economy technologies? (4) What public policy reforms would improve the quality of IPRs in the new economy? © 2003, Elsevier Science (USA).

Appropriability Ability to capture the returns to an investment.

Common good A good or service that is rival and nonexcludable.

Copyright Legal title to a creative expression.

Essential facility An input that several firms depend on for their production activities (such as a bridge, airport runway, or local telephone network).

Patent Legal title to an inventive product or process.

Public good A good or service that is nonrival (where consumption by one person does not diminish its availability for others) and nonexcludable (where nonpayers cannot be prevented from consuming).

Trade secret Private information of commercial value.

Trademark Legal title to a business symbol or mark.

I. INTRODUCTION

The Internet is perhaps the most distinguishing technological phenomenon of the past decade. Whereas the economic impacts of Internet-related innovations are much debated, legal issues are also at the forefront of debate. A core issue is the appropriateness of intellectual property rights (IPRs) over those Internet-related innovations. Economists have long argued that a market system depends on at least two critical ingredients for its smooth functioning: (1) a price mechanism and (2) a system of well-defined property rights. The question is whether property right principles extend to the new economy. De Long and Summers (2001) assert: "It is a fact that we today simply do not know yet how to make the intellectual property system work for the new economy." Though this is a view shared by many, it is also the case that many do not think the current system is broken or inapplicable to the new economy. Hence, the purpose of this chapter is to present some background information and evaluate different perspectives.

It should be noted that debates like this have always been around for new technologies (even in "old economy" industries). When dealing with new uncharted intellectual territory, society often confronts the problem of perhaps granting protection that is too strong or too broad in the beginning, whether it be for automobiles, organic chemicals, therapeutic drugs, computer software, or Internet innovations. In retrospect, after several discoveries have been made, certain legal protections do appear to have been broadly granted. But society does not have the benefit of this perspective when making laws or allocating property rights.

This chapter is outlined as follows: Section II reviews the types of intellectual property rights that can protect new economy innovations. Section

III reviews the state of intellectual property laws pertaining to the new economy in the United States and abroad and reviews some trends in new economy innovations in the United States. Section IV critiques arguments as to whether strong intellectual property rights are appropriate for these new innovations. Finally, Section V critiques policy proposals to improve the intellectual property system for the new economy. By *new economy innovations*, this chapter is referring to software and Internet-related innovations. Telecommunications technologies (such as broadband) or biotechnological innovations will not be treated.

II. INTELLECTUAL PROPERTY RIGHTS

First, before discussing the instruments of IPRs, it is worth reviewing their justification. Knowledge creations have the characteristics of a public good: nonrivalry in consumption and nonexcludability in use. These characteristics create some difficulties. First, they make it easy to freeride on the creative efforts of others. No one has an incentive to pay for something he can get for free. Thus, creators of new knowledge have difficulty appropriating the returns to their investment. Once the sunk costs of knowledge-related investments are incurred, the marginal cost of reproduction and distribution may be low (or zero). Imitators do not have to repeat the investment to sell the same product. With competition and zero economic profits, the creator is unable to recoup the sunk costs. Knowing this, creators have little or no material incentive to undertake the investment and, hence, the argument that, left to its own devices, the private market results in an underinvestment in creative or innovative activity.

Thus, an intellectual property right does the following: it creates excludability for a *temporary* period. No one else can exploit the knowledge during that period. The knowledge creation is still nonrivalrous in nature, but exclusive. Of course, depending on the nature of the knowledge, some natural barriers to the reproduction and distribution of new knowledge may exist (due to the complexity of the innovation or to how the new knowledge is stored, embodied, and accessed, for example, through a book or CD-ROM). Thus, knowledge goods in actuality are nonrivalrous and *partially* excludable. Knowledge creators may also in certain situations have natural advantages (for example, early lead time or reputation). To the extent that these natural barriers and advantages are not adequate, intellectual property rights may help solve the appropriability problem.

Due to the ability of the proprietor of the knowledge good to produce or license its creation exclusively, the price of the product or service can exceed marginal cost for a temporary period of time. The gap between price

and marginal cost enables the creator to earn positive economic profits over that limited period of time, such that the present discounted value of the stream of economic profits could at least exceed or equal the sunk costs of the knowledge investment. Thus, the inefficiency (from a static welfare point of view) is that price does not equal marginal cost. From a dynamic point of view, however, the innovation occurs. The requirement that price equal marginal cost means that society may have to forgo the innovation, hence the argument that society faces a trade-off between intellectual creation and diffusion. Society faces either competitive pricing and possibly no innovation (and therefore little or nothing to diffuse) or monopoly pricing and innovation, which is not diffused or supplied optimally (from a competitive market perspective).

There are additional benefits and costs associated with IPRs. The extra benefits are that, after the temporary period of protection, the knowledge good enters the public domain. It is no longer an exclusive good for some owner, unlike physical property (such as land), which has no automatic termination date. Moreover, in the case of patented technologies, an additional benefit is that the innovator discloses the technological details, so that knowledge spillovers can be generated. Thus, whereas the use of the new idea is restricted, access to the underlying knowledge is not. As for social costs, IPRs can create entry barriers, for example, the royalties or licensing fees that entrants have to pay incumbents. Related to this are the costs of lengthy negotiations and bargaining over licensing rights. IPRs also block others from using a similar expression, technological function, or symbol or word in the same line of business, where by "similar" it is meant similar enough to be infringing. Hence, IPRs can raise the cost of innovation for others. The new creator or innovator must differentiate her product sufficiently to avoid infringing on existing intellectual property rights. In some industries, the permutation of ideas may be limited (for example, words, shapes, and colors), so that the next creator has to "scrape the bucket" to come up with something different.

Given the justification for IPRs, what are the instruments for intellectual property protection? First, patents protect inventions that embody *ideas* of industrial applicability (such as processes, machines, manufactures, and compositions of matter). Traditionally, this meant industrial machines and designs, chemicals, and medicines, but lately genetically engineered life forms, software, and certain methods of conducting business have also qualified for protection. Patents are awarded for inventions that are novel (are not in the public domain), nonobvious (have inventive steps that are not obvious from existing knowledge), and industrially applicable (say, for commercial use). Thus, precluded from patent protection are laws of nature, natural phenomena, and abstract ideas such as basic scientific principles and

mathematical knowledge. Patents last for 20 years from the date of application, but they must be renewed from time to time (or else the right expires).

Second, copyrights protect original forms of *expression*, such as literary, dramatic, musical, and sculptural works, that are fixed in some tangible medium (such as paper, tapes, film, and so forth). Traditionally, this meant books, songs, motion pictures, and the like, but lately sound recordings, computer software, and certain databases (if they involve creative display) have been added to the list of protectible works. Copyright grants the owner the exclusive right to reproduce, distribute, and exhibit the work and to create derivative works. Unlike patents, which protect the substance of ideas, copyrights do not protect ideas.¹ In the event that something is both an idea and an expression (i.e., the only way to express the idea), copyright cannot be provided. Certain expressions also cannot be copyrighted because they are functional (for example, a spreadsheet's menu hierarchy, which is needed to operate a program). Copyrights in the United States are automatic in the sense that they need not be registered. The duration of protection is life plus 70 years (or 95 years for works for hire), which has been increased several times in the past and may be increased again in the future.

Third, trademark rights protect words, symbols, names, devices, or any combination used in commerce that distinguishes one product from others. Trademark rights last for 10 years typically and are renewable upon proof of continued use. If they are not used for some time (or simply hoarded), trademark protection is forfeited. The trademark must also be for commercial use and must describe a good rather than a class of goods like corn flakes or clocks. (This applies to Internet products or services as well, for example, wine.com or bank.com.) These terms are considered generic. Indeed, words or symbols that become generic can no longer be trademarked (or renewed), such as aspirin, escalator, or nylon. Trademarks help protect against misappropriation or tarnishing of a name or symbol. Such protection is deemed to give owners the incentive to invest in both their products and reputation. Trademarks also help protect the public from confusion, assuring the public that they are purchasing what they expect. In places where counterfeit goods are sold, the goods may only have the same identifying mark but not the same quality.

Finally, trade secrets are also an important form of intellectual property rights in the new economy. Trade secret protection applies to any information that provides a business a commercial advantage. The information may be a formula, technique, software source code, or compilation of data (for

¹For example, *Romeo and Juliet* and *West Side Story* are essentially the same idea, but represent different expressions of that idea.

example, customer lists). The owner must have maintained the information in secrecy or disclosed it only under a confidentiality agreement. Trade secrecy rights do not, however, protect against independent discovery or reverse engineering. The trade secret protection is infinite in duration (or until the information falls into the public domain). Protection is also automatic, requiring no formal registration. The holder of the trade secret is merely required to take reasonable steps to keep the information secret.

Now that we have discussed the various types of IPRs and their justification, it would be useful to ask why IPRs have become important in the new economy (other than that they create profitable opportunities for some). These thoughts help to better evaluate the appropriateness of IPRs in the new economy. The explanation stems from the argument that the Internet has changed the nature of ownership of intellectual property. It is much more difficult to exclude others from using, reproducing, altering, and distributing intellectual products that are digital in nature, making copyrights and patent rights especially difficult to enforce. It also makes it harder to manage trade secrets in a world where valuable information takes the form of electronic data and software technology. In the physical world, some costs of reproduction and distribution exist as long as knowledge is embodied in tangible media, such as disks, tapes, films, and paper. Through the Internet, however, it is virtually costless to copy, download, and share electronic copies of infringing material. Moreover, the quality of the electronic or digital product is not compromised through copying or sharing. The practical difficulty of getting users to pay for electronically or digitally based goods thus reduces the ability of creators to appropriate their investments.

The Internet has also increased interest in trademark issues. Previously, quite similar trademarks from different geographic regions could coexist without causing the owners or the public much, if any, confusion. But with the Internet's international coverage, there are greater chances for conflict. Moreover, digitally based counterfeit products will have not just the same mark but the same quality. The Internet has also created new forms of infringement, for example, *cybersquatting*, which involves the registration of a domain name that is based on someone else's trademark, the purpose of which is to extort royalties from the owner and/or to divert Web traffic toward its own business. In fact, most Internet trademark litigation has been over domain name piracies of trademarks. Other new forms of infringement include *framing*, whereby the alleged infringer allows Web users to view the contents of another Web site inside a window or frame within the infringer's own Web site, and *linking*, whereby the alleged infringing Web site provides a link to information at another site, enabling Web users to bypass the home

page of the proprietor.² This gives the appearance that the information belongs to the infringer's Web site.

Not only has the Internet affected intellectual property, but IPRs are also changing the way the Internet operates. The development of the Internet began in an open venture spirit, through the joint efforts of many. The original inventors did not stake out formal intellectual property claims. Instead, a culture of sharing and cooperation had dominated. Moreover, the early development of the Internet was publicly funded. Presently, the Internet is much more commercialized and has moved beyond its early roles as an input into academic research and as a communications medium in public service (for example, in the defense sector). The World Wide Web phenomenon, for instance, has turned the Internet into a marketplace. The next section turns to a key development in the commercialization process: the establishment of property rights.

III. TRENDS IN LAWS AND INTERNET IPRs

In 1976, the U.S. Copyright Act was amended to allow computer programs to be protectible as a literary work. It was not until the early 1980s that computer software could be patentable. In a 1972 case (*Gottschalk vs Benson*), the court system rejected a computer software patent, arguing that the operations of the program were algorithms and thus not patentable. However, in a 1981 ruling (*Diamond vs Diehr*), the court affirmed that software is patentable so long as the underlying algorithms are incorporated in such a way as to produce an industrially applicable product or process. For example, the *Diehr* invention concerned a method of curing rubber. The software program was not an end in itself but a means, just as the laws of physics are involved in a machine patent.

Another relevant court ruling comes from the 1998 case of *State Street Bank vs Signature Financial*. This appeals court ruling is widely hailed as opening the doors to what are known as business method patents. Prior to this ruling, methods of conducting business were not considered patentable subject matter. The 1998 ruling suggests that a business method or model can produce a tangible result and therefore should not be precluded from consideration as an inventive matter. In this instance, Signature Financial owned a patent to a "hub-and-spoke" method for mutual fund valuation. Mutual funds (spokes) pool their assets into one investment portfolio (hub) and a computer program calculates the net asset value of the mutual funds.

²An example of such a trademark dispute is *Ticketmaster vs Microsoft*, where the latter created a deep link into a ticket ordering page on Ticketmaster's site.

This ruling has also led to the growth of financial patents, as documented in Lerner's (2000) study. Indeed, several investment banks own financial patents, including Merrill Lynch Inc. for the method of computing the tax implications of investment transactions. Two economists, one of whom is a Nobel Laureate (Franco Modigliani), own a patent for the discovery of a tax loophole. They devised a method of avoiding taxes by using a credit card to borrow money from a 401K pension fund.

These rulings on computer software and business method innovations helped pave the way for Internet intellectual property activity. One reason for this is that many technologies of the Internet are computer- and software-based. Another reason is that the Internet gave rise to new kinds of firms, namely, e-commerce businesses. These businesses sought intellectual property protection on the grounds that they created "novel" business models, including new methods of advertising, shopping, trading securities, delivering goods and services, and entertaining consumers.

Similar developments have occurred outside the United States. Through a series of legislative reforms from 1993 to 1999, Japan allowed for the patentability of software and business methods that utilize computer hardware for solving problems. In November 2000, the European Union deleted the "computer program exception" to patentable subject matter [in Article 52(2c) of the European Patent Convention Revisions]. The World Trade Organization's Trade Related Intellectual Property Rights (TRIPS) Agreement of 1994 requires signatory nations to protect computer programs as literary works through copyrights and to extend trade secret protection generally, but it leaves open the patentability of software. Given its time frame, the TRIPS agreement does not address Internet issues. Instead, various treaties of the World Intellectual Property Office (WIPO) have dealt with IPRs and e-commerce. One important treaty is the WIPO Copyright Treaty of 1998, which was implemented into U.S. law through the Digital Millennium Copyright Act (DMCA) in 1998. The DMCA represents a major revision of U.S. copyright laws for the digital era.³

Table 1 shows the state of software protection across selected countries and helps give a sense of how widespread such protection has become. The table indicates whether the different kinds of intellectual property rights for software are available as of the end of 1999. "Yes" indicates that protection is available, "no" indicates that it is not, and "maybe" indicates that

³A full discussion of the WIPO treaties and DMCA, however, is beyond the scope of this chapter. Intense debate surrounds their provisions, one of which is the *anticircumvention provision*, which makes it illegal to create technologies that circumvent copyright encryption systems. Critics would contend that this provision limits the exercise of *fair use* (i.e., the right, under certain conditions, to reproduce works without the permission of the copyright holder). Interested readers can obtain further information from www.educause.edu/issues/dmca.html.

Table 1
Software Intellectual Property Rights: An International Comparison^a

Continent	Country	Trade secrecy protection for software	Copyright protection for software	Patent protection for software
North America	Canada	Yes	Yes	Yes
	Mexico	Yes	Yes	Maybe
	United States	Yes	Yes	Yes
South America	Brazil	Yes	Yes	No
	Chile	Maybe	Maybe	No
	Uruguay	Maybe	Yes	No
	Venezuela	Yes	Yes	Maybe
Europe	Austria	Yes	Yes	Yes
	Denmark	Yes	Yes	Yes
	Finland	Yes	Yes	Maybe
	France	Yes	Yes	Yes
Germany	Yes	Yes	Yes	Yes
	Italy	Yes	Yes	Yes
	Netherlands	Yes	Yes	Yes
	Norway	Yes	Yes	Maybe
	Sweden	Yes	Yes	Yes
	Switzerland	Yes	Yes	Yes
	United Kingdom	Yes	Yes	Yes
Asia	India	Yes	Yes	No
	Israel	Yes	Yes	Yes
	Japan	Yes	Yes	Yes
	Korea	Yes	Yes	Yes
	Pakistan	Maybe	Yes	No
	Singapore	Yes	Yes	Maybe
Oceania	Australia	Yes	Yes	Yes
	New Zealand	Yes	Yes	Yes
Africa	Kenya	No	Maybe	No
	Mauritius	No	No	Maybe
	South Africa	Yes	Yes	No

^aYes (no) indicates that statutory protection is (is not) available. Maybe indicates that protection may exist depending on government decree and court decisions. Source: Louwers and Prius (1999).

protection is ambiguous, depending on court rulings and government decrees. Most of the countries listed provide copyright protection for software. Mauritius does not provide copyright protection for software, whereas Chile and Kenya provide ambiguous protection. Most countries also provide trade secrecy protection for software, with the exception of Kenya

and Mauritius. Relatively speaking, patent protection for software is not as widely available as the other means of software protection. Most European countries, the United States, Canada, Australia, New Zealand, Japan, and Korea allow software to be patented. However, many developing countries do not recognize software as patentable, or at least do not provide formal statutory recognition. Sixteen nations allow for all three forms of protection for software (copyrights, patent rights, and trade secrecy), including the United States, Canada, Japan, Israel, and the United Kingdom.

How significant software innovative activity has been and whether the new laws have stimulated software innovation are difficult to determine due to data problems. One way to measure the intensity of software innovative activity would be to examine patenting and copyright activity. Because copyright is automatic (not requiring any formal registration), the data on copyright registrations do not capture the actual quantity of new creative works. The problem with patent data is that, as of yet, no separate patent class for software has been created for statistical purposes. Instead, when patents are awarded for software-related innovations such as new data structures or information retrieval methods, the inventions are recorded in the traditional invention classes that most correspond, for example, data processing (class 700) or information processing (class 395). On the basis of these two invention classes, Cowan and Harison (2001) infer that software patenting increased from 100 patents in 1980 to 1600 in 2000. Greg Aharonian of Internet Patent News Service identified an additional 19 possible invention classes into which software patents can be assigned and found that there were 1050 software patents in 1980 and 15,743 in 2000. Thus, both estimates, though differing in levels, suggest that the growth in software patenting over the past two decades was about 16-fold.

Internet-related patents are, however, assigned a separate class. Table 2 shows a sample of some Internet patenting statistics in the United States. Currently, Internet innovations account for a very small percentage of total U.S. patents granted, but they account for a growing share during 1993–2000. Part A of Table 2 shows the number of patents awarded for Internet-related innovations. Part B shows how many of these Internet patents were for business methods. Just under 10% of Internet-related patents awarded in 1997–2000 were for business method innovations.

Internet-related innovations in general accounted for 0.045% of all patents granted in the United States in 1993 (0.04% were accounted for by U.S. residents and 0.005% by nonresidents). By 2000, 3.31% of all U.S. patents were awarded for Internet-related innovations (2.7% were accounted for by U.S. residents and 0.61% by nonresidents). Thus, between 1993 and 2000, a 100-fold increase in Internet patenting occurred, more than

Table 2
Internet Patenting Activity^a

	Number of patents granted		
	1993	1997	2000
A. Internet-Related Innovations (Total)			
U.S. residents	49 (0.04%)	560 (0.45%)	4679 (2.7%)
Foreign residents	5 (0.005%)	99 (0.08%)	1066 (0.61%)
Japan	3	50	468
Canada		14	141
United Kingdom		9	76
Israel		4	50
France	1	5	44
Germany		2	39
Korea	1	1	30
B. Internet-Related Innovations (Subclass: Business Methods)			
U.S. residents	3 (0.002%)	51 (0.04%)	457 (0.26%)
Foreign residents		5 (0.004%)	70 (0.04%)
Japan		2	35
Canada		1	10
United Kingdom		1	7
Germany			4
Israel			4
Finland			2
Sweden			1
Netherlands		1	0

^aFigures in parentheses indicate the percentage share in total U.S. patents granted in that year. Source: U.S. Patent and Trademark Office (2001).

in any other field. Americans have by far been the most active Internet patentees. The next most active are the Japanese, followed by the Canadians and British. Internet patents have been awarded for communication protocols, data compression methods, interfaces (platforms through which hardware, software, and the user interact), networking techniques, encryption methods, information processing, text searching system, and so forth.

As for Internet business method patents, 3 were granted in 1993 (with all 3 going to U.S. residents) and 56 in 1997 (only 5 of which went to foreigners). By 2000, 527 such patents were awarded, 86% of which went to U.S. residents. These Internet business method patents accounted for 0.3% of all U.S. patents granted in 2000. Japan accounted for half of the business method patents awarded to foreigners. Canada accounted for one-seventh, and the United Kingdom accounted for one-tenth.

It would be useful to provide a few examples of these Internet business method patents as they have stirred much controversy, particularly over how "obvious" they seem. One category of such patents concerns methods of sales. For example, Amazon.com acquired a patent for its one-click shopping method. The idea is that customers would only have to enter their names and billing and shipping information once (that is, they incur a fixed one-time cost of entering information) so that for future purchases they need only click on a single button to complete a purchase. Priceline.com acquired a patent on reverse auctions, where buyers indicate what they want to buy and at what price, and the company helps search for sellers. Walker Digital holds an on-line patent for determining the price of an option to buy an airline ticket and for facilitating the sale and exercise of such options. Linkshare has a patent for the idea of paying someone for referral services (on-line). That is, a Web site gives commissions to another Web site if the latter provides a link to the first Web site, such that the link results in the sale of the products of the first Web site.

Another category of on-line business method patents concerns methods of advertising. Cybergold acquired a patent for the idea of paying customers to view advertisements on its site. Net Zero has a patent on the idea of pop-up advertising windows. Juno has one for the idea of storing ads and other messages while the user is off-line or preoccupied with electronic mail.

Table 3 reports the number of patents awarded in the United States to selected firms during 1996–2000. Part A focuses on Internet-related patents in general, whereas part B focuses on Internet-related business method patents in particular. The table shows that most of the Internet patents have been granted to large, well-established firms, such as IBM, Microsoft, Sun Microsystems, AT&T, Lucent Technologies, and Pitney Bowes. The Amazon.coms, Walker Digitals, and Cybergolds account for a small share, though their patents have received much of the attention. A biased impression therefore might be formed from focusing on the Internet business method patents of a few relatively new entrants. Nevertheless, for *Amazon.com*, 5 out of 8 of its Internet patents are for business methods; for *Walker Digital*, it is 5 out of 13. For the top 5 Internet patenting firms, less than 10% of their Internet patents are business-method-related.

Table 4 provides some data on Internet-related trademark filing activity. Data are not as widely available as for patenting activities. In 1994, there were 307 trademark filings pertaining to Internet-related goods and services. This accounted for 0.2% of all trademark applications that year. Twenty-six of these designated ".com" as part of their trademark. The rest were for general Internet-related goods and services, such as logos in World Wide Web businesses, or for products and services sold through the Internet (e.g., cosmetics, insurance, and management consulting services). By

Table 3
Internet Patenting—Leading Firms^a

Company	Number of patents awarded 1996–2000
A. Internet-Related Innovations (Total)	
Top five:	1378
International Business Machines (IBM)	514
Microsoft	451
Sun Microsystems	280
AT&T	278
Lucent Technologies	
Others	167
Hewlett-Packard	116
Sony	39
Netscape	27
British Telecom	13
America Online	13
Walker Digital	8
Amazon.com	3
Qwest Communications	
B. Internet-Related Innovations (Subclass: Business Methods)	
Top five	78
International Business Machines (IBM)	35
Microsoft	27
AT&T	22
Pitney Bowes	18
Sun Microsystems	
Others	13
Lucent Technologies	13
Xerox	10
Hewlett-Packard	5
Amazon.com	5
Walker Digital	4
Priceline.com	3
Merrill Lynch	3
Supermarket Online	2
Cybergold	2
Juno Online	

^aSource: U.S. Patent and Trademark Office (2001).

2000, there were 41,816 Internet-related trademark filings, accounting for 14.5% of all trademark applications that year. Of these, nearly one-fourth designated ".com" as part of their trademark. However, with the dot com bubble bursting in 2001, Internet-related trademark activity decreased significantly that year, thus exhibiting some procyclicality. The decline in

Table 4
Internet Trademark Filing Activity^a

	Number of trademark filings		
	Internet-related goods and services		Computer-related goods and services
	Total	Dot com	
1994	307 (0.2%)	26	
1996	8,212 (4.5%)	764	
1998	12,235 (6.1%)	1,665	
1999	27,575 (12.7%)	12,164	65,072
2000	41,816 (14.5%)	13,412	79,699
2001	16,867 (7.6%)	2,744	40,257

^aFigures in parentheses indicate the percentage share in total U.S. trademark filings for that year. Dot com filings refer to trademark applications where ".com" is part of the trademark. Source: Dechert (2000–2002).

Internet trademark activity was reflective of a general decline in high-tech trademark filing activity. In the computer hardware and software industry, there was also a significant drop in trademark filings in 2001.

IV. APPROPRIATENESS OF IPRs FOR THE INTERNET

Are these trends in Internet-related IPR activity cause for concern? This section discusses some of the concerns and provides some commentary on the conceptions and misconceptions. The issues raised here are by no means exhaustive but should reflect the intensity of debate.

A. ON FUNDAMENTAL KNOWLEDGE

One concern is that IPRs adversely affect progress in basic science. As DeLong and Summers (2001) point out, many past and modern technologies (such as navigation instruments for ships) owe their existence to basic research, including pure mathematics. The authors thus stress the importance of allowing for the broad dissemination of basic science. Barlow (1994) expresses the concern that, throughout history, only the expression of ideas was the property of thinkers. Ideas themselves were the collective

property of humanity, but "we are now claiming to own ideas themselves and not merely their expression. And since it is likewise now possible to create useful tools that never take physical form, we have taken to patenting abstractions, sequences of virtual events, and mathematical formulae." However, as was discussed earlier, the IPR system was not designed to protect fundamental knowledge. Laws do not permit the patenting of basic scientific and mathematical principles, but the patenting of applied scientific results.

However, this is in principle. In practice, particularly in the case of information technologies, the distinction between applied and basic knowledge can be blurred. For example, computer software programs are considered algorithms, and it may be difficult at times to separate the mathematical problem-solving innovations from the "technical effects" produced (i.e., the problems being solved). This difficulty of distinguishing basic vs applied knowledge also arises in biotechnology. Property rights have been granted for *expressed sequence tags* (ESTs), which are fragments of genes and which critics have claimed are essentially research *tools* that scientists use to profile gene expressions. Dreyfuss (2000) likewise argues that business method innovations are more *basic* than applied. Business methods, she argues, are akin to sport moves. They are fundamental to the very activity in question (much like ESTs are to the scientific researcher).

B. PUBLIC DOMAIN CONCERNS

Specifically, the concern is that IPRs are allowing firms to claim proprietary rights to knowledge that is in the public domain. David and Foray (2001) comment that, "a proliferation of intellectual property rights inhibits access to such information in areas (basic research in general, life sciences, software) where new knowledge had remained largely in the public domain." In addition, David and Foray (2001) argue that the problem "especially for the long run, is the continuing encroachment upon the domain of public information through the efforts to find new ways to privatize old knowledge." Again, one must be careful here because the laws do not permit the patenting of ideas that are already in the public domain, for they fail the test of novelty. It is correct to state that IPRs prevent technologies that otherwise would have gone into the public domain, but not that IPRs will convert technologies already in the public domain into private property.

On the other hand, a legitimate concern is that few technologies are (eventually) entering the public domain. For example, the duration of copyright is so long—and current owners have had such success at lobbying

politicians for extensions in copyright life—that protection appears to last infinitely. Moreover, Internet and computer technologies have such short life cycles that technological obsolescence occurs well before the expiry of an intellectual property right. In effect, there is no time period during which the public gets to use new knowledge for free because, by the time it is in the public domain, the technology is obsolete.

C. COMMONS ARGUMENT

According to Lessig (2001), “creativity flourished . . . because the Internet protected an innovation commons.” The Internet was and could still be a neutral platform upon which to innovate and disseminate. Yet the Internet seems to be transforming from an open place for ideas into an industry modeled after cable television. Once core technologies are removed from a commons area, innovators are no longer free to build upon them as they wish. The view here is that the Internet grew precisely because the resources were not divided up among private owners. Lessig (2001) suggests that enclosure of the innovation commons may spell the demise of the Internet. With stronger IPRs, the Internet will no longer provide a neutral platform for all potential users or innovators.

In addition, there is also an anticommons argument developed by Heller and Eisenberg (1998), which suggests that knowledge will not be used enough if knowledge becomes very fragmented and if many different proprietors own the many different fragments. Knowledge ends up being underutilized because of the high transaction costs of negotiating cross-licenses among the many private parties. Wu (2001) applies the anticommons principle to the Internet economy to argue that, “just as highly divided ownership of a common property can hinder profitable exploitation of that property, . . . highly divided ownership of common network components will hinder the development of the network itself.”

These are serious potential consequences. But first some clarifications on terminology. Commentators are not using the term *common property* conventionally, which can create confusion and do a disservice to their main argument. A common resource is rivalrous in use but not excludable, largely because of the absence of property rights. For common goods, property rights are typically a solution to preventing a “tragedy of the commons.” Digital information and other Internet-related knowledge goods are public goods, not common. They are not rivalrous goods. The commons paradigm would suggest that resources would be depleted through overuse, which is not the case with information goods. For example, the same source code or electronic document can be used over and over again without diminishing

the supply. As for the anticommons paradigm, it is not clear to speak of a “divided ownership of common property.” Once property is owned it is not common and should not be addressed as such; second, once owned, the intellectual property is still nonrival. The more appropriate paradigm is that of a club good. The good is still nonrivalrous but excludable. Only members of the club can exploit the good (namely, the proprietor, its licensees, and paying customers).

But to address the critics’ concerns, what they have in mind is that the resources (the innovations) were initially free goods. More importantly, they served as a *shared* infrastructure among users. The problem is not so much that there is a charge for the good (though this of course reduces consumer surplus), but rather that these innovations constitute *essential facilities* (which both the firm and its rivals need to operate) and that private ownership of them creates *bottlenecks*, in much the same way that bottlenecks are created when one railroad operator owns a mountain pass that other railroad companies need to go through or when a phone company owns a local telephone network that other companies need to access. The proprietor has scope to behave anticompetitively, providing access only to members of the “club.” In this case, it is useful to distinguish between competition law and intellectual property law. The problem (and its resolution) may fall more within the scope of competition policies than that of IPRs. Instead of limiting IPRs generally, the limitations could be selective. Because not all Internet innovations are bottlenecks, competition policies could focus on those technologies that are.

D. NETWORK EFFECTS ARGUMENT

Positive network effects occur when the value of a good increases with the *number of users* of the good (consider, for example, a telephone system). Network effects work best when a common technological standard exists through which members of the network can more easily interact. When many technological parts are involved, it helps if they are all *interoperable* or compatible so that a standard at large can be established.⁴ A major consequence of network effects is that a particular design or system tends to emerge as a *de facto* standard. To see this, suppose that there are two competing technologies and that one of them has an extra user. This makes that technology marginally more valuable than the other, enough to attract the

⁴For example, in the case of plumbing, the benefits of interoperability can be seen from the fact that sinks, faucets, pipes, and screws can be purchased from different vendors or manufacturers, and yet the products can all connect to one another.

next marginal user. The addition of the marginal user further raises the marginal value of adopting that technology. The process repeats until that technology eventually dominates the marketplace.

Two problems, however, arise due to network effects. First, the market may converge on an inferior technological standard. This may happen if the number of users of the technology more than compensates for its technological inferiority. When convergence takes place, a situation of *lock-in* tends to develop because it is very difficult for users to switch to another standard. Second, a monopoly or oligopoly may result, particularly if there is only one or a few providers of that standard. The existing standard acts as an entry barrier. A new entrant with a different standard needs either a sufficiently higher quality technology or a sufficiently lower price to persuade users to switch. If switching costs are too high (for example, the costs of learning a new keyboard), users may not switch even if the other good is free.

Commentators suggest the need to modify intellectual property laws in industries where network effects arise (such as the software and Internet industries). Network effects plus IPRs may grant excessive market power to the proprietor of a technological standard. Intellectual property rights give extra time for the proprietor to exploit network economies. Moreover, because of the network effects, the proprietor has market power even after the intellectual property right expires (assuming the technology is not obsolete).

Where network effects are not especially strong and the tendency for lock-in is weak, intellectual property protection would work to reinforce the lock-in effect [as argued by Cowan and Harison (2001)], which may be costly if the protected technology later turns out to be suboptimal. It may also be costly if the protected technology later turns out to be statutorily invalid (for example, an invalid patent). As Dreyfuss (2000) argues, the invalidation is of no consequence if lock-in causes the public to become accustomed to that technology.

In other situations, IPRs may reduce the benefits of networks by making it difficult to achieve interoperability. This would occur if the various technological components that comprise a network system are owned by different proprietors and if these proprietors block one another from using their particular technology (say, by refusing to grant a license at an affordable fee). In this case, the standard setting process would break down.

The following are some criticisms of the perspectives on IPRs suggested by network effects. First, it should be clarified that, even without IPRs, a technology may become a standard. Second, evidence of lock-ins and/or of convergence to an inferior standard is mixed. Researchers have disagreed on the interpretation of case study histories (for example, the Qwerty vs

Dvorak keyboard). Third, not all protected technologies are radical innovations. Varying degrees of quality exist. Some are very inventive, whereas others are minor. The distribution is not uniform, but skewed; a minority of inventions represent major breakthroughs. Thus, not all inventions have the potential to become standards or, as discussed earlier, to constitute essential facilities.

Fourth, the discussion thus far has focused on the demand side (the users who join a network) but neglected the supply side. For a monopoly provider of the standard to emerge, network effects are not sufficient. Network effects affect the demand for membership in the network. But unless the supply side has the capacity to accommodate more members and produce more goods, the demand side will be frustrated. For example, if a provider faces *rising average costs of production*, that is, if its total production costs per unit of output increase as it expands production, it is not likely to supply the market on its own. The market would require multiple firms to satisfy demand. Hence, not all industries where network effects exist will necessarily be supplied by a monopolist. This might explain why in the Internet market several service providers can coexist (for example, AOL, AT&T, CompuServe, Earthlink, and MSN).

E. VALIDITY ISSUES

Perhaps the strongest criticism of some software and Internet-related innovations is not that they are common goods or network goods, but that they are not statutorily valid. They did not satisfy the requirement of either nonobviousness or novelty. Critics point out (for example, in the case of Amazon.com's one-click patent and Priceline's reverse auction patent) that the "inventive steps" are very small and that much prior art is available. Prior art refers to previously existing knowledge (in previous patents, dissertations, scientific journals, trade publications, and so forth). Often it appears that Internet business method ideas are simply borrowed from the traditional (bricks-and-mortar) world and adapted to cyberspace. The issue is whether this type of adaptation constitutes sufficient novelty.

V. POLICY REFORMS

What are the policy lessons in light of the previous discussions? This section raises more issues than it settles. Its focus is on suggesting factors for consideration rather than making recommendations. By way of summary, some of the main concerns about IPRs in the new economy are

the needs to (1) better protect the marketplace from bottlenecks; (2) ensure that fundamental knowledge remains in the public domain; and (3) avoid granting IPRs to technologies for which prior arts exist and for which the inventive steps are small.

A. DEALING WITH BOTTLENECKS

For standard setting technologies or technologies that serve as essential facilities, some limitations on IPRs are necessary, whether the technologies are operating standards, computer languages (C++), interfaces (Windows), or protocols (TCP/IP). Some lessons from trademark law may be relevant. If a symbol or mark becomes generic, protection is denied (or not renewed). Likewise, with copyrights, if an expression is functional or an industry standard, protection cannot be provided. This has been the case with interface specifications (such as a computer's central processing unit, memory, and operating system). Through court rulings, interface specifications have gradually been declared noncopyrightable.⁵ As of yet, standard setting exceptions to patentability have not been legislated. Alternatively, the private market could be allowed to try to resolve essential facilities and standard setting issues through private negotiations. An international example of where private bargaining worked is the 3G (third generation) wireless standard. If private negotiations do not work and blocking or withholding still exists on the part of some firms, the public authority or an arbitration board could consider ordering *compulsory* licensing. The drawback of such an order is the possible adverse effects on future innovation.

B. PRESERVING FUNDAMENTAL KNOWLEDGE IN THE PUBLIC DOMAIN

As simple as it sounds, this is notoriously difficult when dealing with new technological fields. It is sometimes difficult to judge *ex ante* what is scientific principle and what is scientific application. Basic knowledge must be preserved in the public domain not only to allow applications to be derived therefrom but also to further advance basic knowledge. One practical consideration is to shorten the duration of intellectual property protection for new, uncharted technological fields (which is where most of the granting problems seem to occur) and to lengthen the duration as the fields mature. A shorter duration would also be beneficial in situations where technolo-

⁵For an analysis of the legal cases, see Samuelson (1995).

gies have short life cycles. The disadvantage is that short durations may provide inadequate incentives for pioneering innovations.

C. REDUCING THE INCIDENCE OF INVALID PATENTS

This requires weeding out patent applications that are neither novel nor nonobvious. To better spot innovations that are not novel, patent examiners need better information. This requires building up society's prior art database. Of course, over time, the lack of prior art information is somewhat taken care of naturally as a technological field matures and as more research is conducted and discoveries are made that are added to society's knowledge base. But during the transition, invalid patents may (or will) be granted that impose costs on society. Thus, society needs to invest resources in developing this database, which would include both patented prior art and nonpatented prior art (based on scientific journals, social practices, and so forth).

Another way to obtain more prior art is simply to ask the private sector (particularly the rivals of the firm applying for a patent). It would be in the interest of rivals to come up with evidence of prior art. Rivals could also enlist the aid of others by offering a reward. For example, an Internet firm known as Bounty Quest has emerged whose primary business is to help clients offer a bounty to those who can find prior art to an existing patent. Clients who are likely to use this service are those involved in litigation, among other things. The clients get others to assist in the search for prior art by offering a monetary reward. Thus far, a number of weak Internet patents have been discovered as a result of this business. Interestingly, the business of Bounty Quest itself was granted an Internet business method patent.

A reduction in the incidence of invalid patents also requires that patent examiners devote more examination time per application. The average time spent per patent application has decreased over the years. Spending more time does not guarantee increased *quality* time, and thus Merges (1999) suggests using incentive mechanisms such as an employee performance bonus system to reward examiners with low "error rates" (that is, low rates of invalid patents granted).

Another policy is to increase the *nonobviousness* standard. But this is much easier said than done. Many consider the nonobviousness requirement as the ultimate test of patentability. Is there more than a trivial technical achievement? How should this be determined? One suggestion is to define inventive step in terms of economic value. Merges (1999) discusses the proposal of introducing economic information: "[I]s there any way for

the [Patent Offices] to sort patent applications by anticipated economic value?" Merges then provides a blueprint for how this might be done. If patent offices and the courts do incorporate economic information, this would be a large change in practice. Until now the legal systems around the world have been judging inventions on the basis of three criteria: novelty (is there prior art?); nonobviousness (is the idea not obvious to a person having ordinary skill in the art?); and industrial applicability (is there, say, a commercial use?). None of these criteria explicitly requires anticipating the economic value of an invention. Thus, a minor innovation (such as a new paper clip) can be patented if it meets those three criteria.

Whereas the addition of economic information may be a good idea, there are a number of difficulties to overcome. The first set of difficulties is in deriving market value, especially for a brand new idea, product, or process. The market value of an invention is likely to be path-dependent (dependent on the company of other inventions around or to arise). The market value is also likely to depend on extrinsic factors like managerial ability, marketing, consumer habits, tastes, incomes, taxes, and monetary and trade policies. A second set of difficulties is information. The proposal to incorporate economic information requires the examiner to have business and industry data. In requiring applicants to provide cost information, patent offices may face a moral hazard whereby applicants inflate their cost figures (in order to make it seem that patent protection is needed to help recoup their investments). The patent examiner will also require information about the relevant industries to which the invention will have value. This would require rather complex input-output tables to figure out where inventions come from and where they go.

An alternative to anticipating the economic value of inventions is to do this indirectly by using the *price mechanism*, namely, to raise the price of acquiring and maintaining patent protection. Because the economic value of inventions is largely unobservable to the patent office, the advantage of raising patent application fees and renewal fees is that the price schedule would let patent applicants and owners self-select. Those who own high-quality inventions will apply for patents and renew their patent rights as long as their inventions have sufficiently high value. Those who do not own high-quality patents may be discouraged from seeking patent rights based on the cost of applying, and those whose inventions have fallen in value will not renew their rights if their inventions are not worth the cost of renewal. In these cases, the market helps determine the value of the inventions. Thus, varying the price of patents might be a useful way to screen for low-quality patents. In addition, as higher fees reduce the volume of patent applications, patent examiners may be able to devote more time per application. This scheme, however, may not fully preempt invalid patents from being

submitted. An innovation that is obvious, not very novel, and basic rather than applied could very well have high private value to a firm but low social value.

Thus, in closing, the pursuit of high patenting standards remains a challenge. Most of the policy proposals, though, seem disproportionately concerned with the phenomenon of Internet-related business method patents. But as Section III pointed out, these patents account for a very small share of total patenting activity. Future policy reforms therefore should be based on a more extensive review of innovative and creative sectors.

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